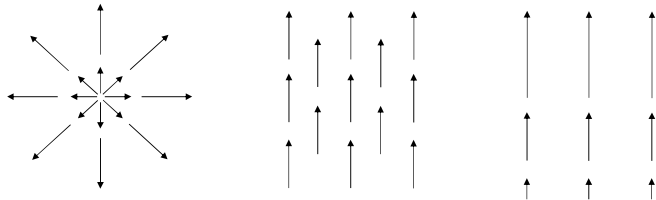


Curl: Example



Home Work: Calculate curl for each field from first principle

1

Important formulae with gradient/divergence/curl

$$\vec{\nabla} \equiv \hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z}$$

is a vector operator – a hybrid entity satisfying two sets of rule:

- (i) vector rule
- (ii) partial differentiation rule

$$\vec{A}(\vec{\nabla} \cdot \vec{B}) \neq (\vec{A} \cdot \vec{\nabla})\vec{B}$$

$$\vec{\nabla}(f + g) = \vec{\nabla}f + \vec{\nabla}g$$

$$\vec{\nabla} \cdot (\vec{A} + \vec{B}) = \vec{\nabla} \cdot \vec{A} + \vec{\nabla} \cdot \vec{B}$$

$$\vec{\nabla} \times (\vec{A} + \vec{B}) = \vec{\nabla} \times \vec{A} + \vec{\nabla} \times \vec{B}$$

2

Important formulae with gradient/divergence/curl

$$\vec{\nabla}(fg) = f \vec{\nabla}g + g \vec{\nabla}f$$

$$\vec{\nabla}(\vec{A} \cdot \vec{B}) = (\vec{A} \cdot \vec{\nabla})\vec{B} + (\vec{B} \cdot \vec{\nabla})\vec{A} + \vec{A} \times (\vec{\nabla} \times \vec{B}) + \vec{B} \times (\vec{\nabla} \times \vec{A})$$

$$\neq \vec{A} \vec{\nabla} \cdot \vec{B} + \vec{B} \vec{\nabla} \cdot \vec{A}$$

$$\vec{A} \times (\vec{\nabla} \times \vec{B}) = \vec{\nabla}(\vec{A} \cdot \vec{B}) - (\vec{A} \cdot \vec{\nabla})\vec{B}$$

$$\vec{B} \times (\vec{\nabla} \times \vec{A}) = \vec{\nabla}(\vec{A} \cdot \vec{B}) - (\vec{B} \cdot \vec{\nabla})\vec{A}$$

$$\vec{\nabla}(\vec{A} \cdot \vec{B}) = (\vec{A} \cdot \vec{\nabla})\vec{B} + \vec{A} \times (\vec{\nabla} \times \vec{B})$$

$$\vec{\nabla}(\vec{A} \cdot \vec{B}) = (\vec{B} \cdot \vec{\nabla})\vec{A} + \vec{B} \times (\vec{\nabla} \times \vec{A})$$

$$\vec{\nabla} \cdot (f\vec{A}) = f \vec{\nabla} \cdot \vec{A} + \vec{\nabla}f \cdot \vec{A}$$

$$\vec{\nabla} \cdot (\vec{A} \times \vec{B}) = \vec{B} \cdot (\vec{\nabla} \times \vec{A}) - \vec{A} \cdot (\vec{\nabla} \times \vec{B})$$

$$\vec{\nabla} \times (f\vec{A}) = f \vec{\nabla} \times \vec{A} + \vec{\nabla}f \times \vec{A}$$

$$\vec{\nabla} \times (\vec{A} \times \vec{B})$$

Home Work

3

Double derivative of a vector function

Applying $\vec{\nabla}$ twice, we can construct second derivative

Case 1 $\vec{\nabla} \phi$

$$\text{Divergence of gradient } \vec{\nabla} \cdot \vec{\nabla} \phi = \left(\hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z} \right) \cdot \left(\hat{x} \frac{\partial \phi}{\partial x} + \hat{y} \frac{\partial \phi}{\partial y} + \hat{z} \frac{\partial \phi}{\partial z} \right)$$

$$= \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2}$$

$$\nabla^2 \equiv \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

Laplacian operator

Curl of gradient $\vec{\nabla} \times \vec{\nabla} \phi = 0$ $\vec{\nabla} \phi$ is irrotational

$$= \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \partial\phi/\partial x & \partial\phi/\partial y & \partial\phi/\partial z \end{vmatrix} = \hat{x} \left(\frac{\partial}{\partial y} \frac{\partial \phi}{\partial z} - \frac{\partial}{\partial z} \frac{\partial \phi}{\partial y} \right) - \hat{y} \left(\frac{\partial}{\partial x} \frac{\partial \phi}{\partial z} - \frac{\partial}{\partial z} \frac{\partial \phi}{\partial x} \right) + \dots$$

$$\frac{\partial^2 \phi}{\partial y \partial z} = \frac{\partial^2 \phi}{\partial z \partial y}$$

$$\frac{\partial^2 \phi}{\partial x \partial z} = \frac{\partial^2 \phi}{\partial z \partial x}$$

4

Double derivative of a vector function

Applying ∇ twice, we can construct second derivative

Case 2 $\vec{\nabla} \cdot \vec{A}$

Gradient of divergence $\vec{\nabla}(\vec{\nabla} \cdot \vec{A}) \neq (\vec{\nabla} \cdot \vec{\nabla})\vec{A}$

Case 3 $\vec{\nabla} \times \vec{A}$

Divergence of curl $\vec{\nabla} \cdot (\vec{\nabla} \times \vec{A}) = 0$ $\vec{\nabla} \times \vec{A}$ is solenoidal

Curl of curl $\vec{\nabla} \times (\vec{\nabla} \times \vec{A}) = \vec{\nabla}(\vec{\nabla} \cdot \vec{A}) - \nabla^2 \vec{A}$

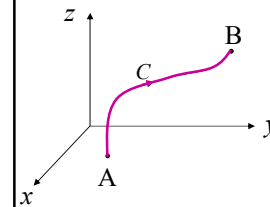
5

Vector Integration

In ED, we encounter several different kinds of integrations

- (i) line integral
- (ii) surface integral
- (iii) volume integral

Line Integral



$$d\vec{l} = dx \hat{x} + dy \hat{y} + dz \hat{z}$$

$$\int_C \phi d\vec{l} \quad \boxed{\int_C \vec{V} \cdot d\vec{l}} \quad \int_C \vec{V} \times d\vec{l}$$

C : contour $\rightarrow$ open contour

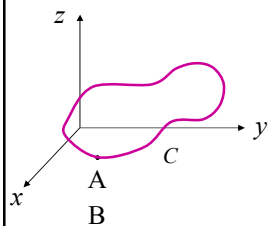
6

Vector Integration

In ED, we encounter several different kinds of integrations

- (i) line integral
- (ii) surface integral
- (iii) volume integral

Line Integral



$$d\vec{l} = dx \hat{x} + dy \hat{y} + dz \hat{z}$$

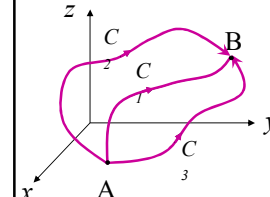
$$\int_C \phi d\vec{l} \quad \boxed{\int_C \vec{V} \cdot d\vec{l}} \quad \int_C \vec{V} \times d\vec{l}$$

C : contour $\rightarrow$ open contour
 $\rightarrow$ closed contour $\oint_C \vec{V} \cdot d\vec{l}$

7

Vector Integration

Line Integral



$$\int_C \vec{V} \cdot d\vec{l} \quad \text{In general path dependent}$$

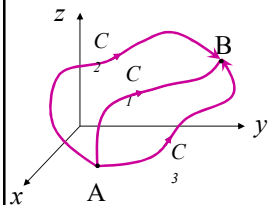
We must define direction of traversal around the loop

Anti-clockwise

8

Vector Integration

Line Integral

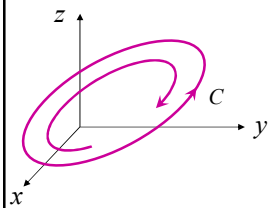


$$\int_C \vec{V} \cdot d\vec{l} \quad \text{In general path dependent}$$

We must define direction of traversal around the loop

Anti-clockwise

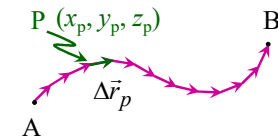
$$\int_C \vec{V} \cdot d\vec{l} \quad \text{changes its sign}$$



9

Vector Integration

Line Integral



$$\int_C \vec{V} \cdot d\vec{l} = \lim_{N \rightarrow \infty} \sum_{p=1}^N \vec{V}(x_p, y_p, z_p) \cdot \Delta \vec{r}_p$$

There exists a class of vector field for which line integral is independent of the path followed $\rightarrow$ conservative vector field

10

Vector Integration

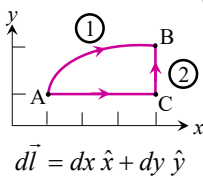
Line Integral: Example

$$\int_C \vec{V} \cdot d\vec{l} = ? \quad \text{where } \vec{V} = (x+y)\hat{x} + (y-x)\hat{y}$$

along (i) parabola $y^2 = x$ from (1,1) to (4,2)

(ii) $y = 1$ from (1,1) to (4,1) followed by $x = 4$ from (4,1) to (4,2)

(iii) $\oint_C \vec{V} \cdot d\vec{l}$ for the loop ACBA



$$\begin{aligned} \text{(i)} \quad I &= \int_{(1,1)}^{(4,2)} ((x+y)\hat{x} + (y-x)\hat{y}) \cdot (dx\hat{x} + dy\hat{y}) \\ &= \int_{(1,1)}^{(4,2)} (x+y)dx + (y-x)dy \end{aligned}$$

but $y^2 = x \rightarrow 2y dy = dx$ & $y: 1 \rightarrow 2$

$$I = \int_{y=1}^{y=2} (y^2 + y) 2y dy + (y - y^2) dy = 34/3$$

11

Vector Integration

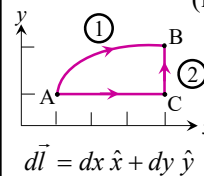
Line Integral: Example

$$\int_C \vec{V} \cdot d\vec{l} = ? \quad \text{where } \vec{V} = (x+y)\hat{x} + (y-x)\hat{y}$$

along (i) parabola $y^2 = x$ from (1,1) to (4,2)

(ii) $y = 1$ from (1,1) to (4,1) followed by $x = 4$ from (4,1) to (4,2)

(iii) $\oint_C \vec{V} \cdot d\vec{l}$ for the loop ACBA



(ii) Along the horizontal segments, $y = 1, z = 0, x: 1 \rightarrow 4$

$$\rightarrow dy = dz = 0 \quad d\vec{l} = dx\hat{x}$$

$$\vec{V} \cdot d\vec{l} = (x+y)dx = (x+1)dx$$

$$I_1 = \int_{x=1}^{x=4} (x+1)dx = 21/2$$

12

Vector Integration

Line Integral: Example

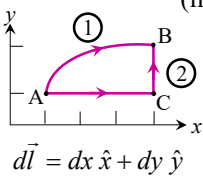
$$I_1 = 21/2$$

$$\int_C \vec{V} \cdot d\vec{l} = ? \quad \text{where } \vec{V} = (x+y)\hat{x} + (y-x)\hat{y}$$

along (i) parabola $y^2 = x$ from (1,1) to (4,2)

(ii) $y = 1$ from (1,1) to (4,1) followed by $x = 4$ from (4,1) to (4,2)

(iii) $\oint_C \vec{V} \cdot d\vec{l}$ for the loop ACBA



(ii) Along the vertical segments, $x = 4, z = 0, y:1 \rightarrow 2$

$$\rightarrow dx = dz = 0 \quad d\vec{l} = dy \hat{y}$$

$$\vec{V} \cdot d\vec{l} = (y-x)dy = (y-4)dy$$

$$I_2 = \int_{y=1}^{y=2} (y-4) dy = -5/2$$

$$I_1 + I_2 = 8$$

13

Vector Integration

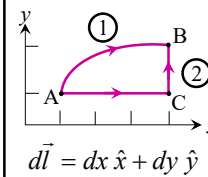
Line Integral: Example

$$\int_C \vec{V} \cdot d\vec{l} = ? \quad \text{where } \vec{V} = (x+y)\hat{x} + (y-x)\hat{y}$$

along (i) parabola $y^2 = x$ from (1,1) to (4,2)

(ii) $y = 1$ from (1,1) to (4,1) followed by $x = 4$ from (4,1) to (4,2)

(iii) $\oint_C \vec{V} \cdot d\vec{l}$ for the loop ACBA



(iii) Along path "1", $\int_C \vec{V} \cdot d\vec{l} = 34/3$

Along path "2", $\int_C \vec{V} \cdot d\vec{l} = 8$

$$\rightarrow \oint_C \vec{V} \cdot d\vec{l} = 8 - 34/3 = -10/3 \neq 0$$

14

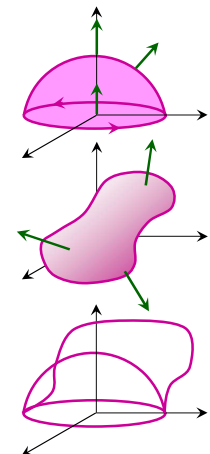
Vector Integration

Surface Integral:

$$\int_S \phi d\vec{a}$$

$$\int_S \vec{V} \cdot d\vec{a}$$

$$\int_S \vec{V} \times d\vec{a}$$



S : surface element $\rightarrow$ open element
 $\rightarrow$ closed element $\oint_S \vec{V} \cdot d\vec{a}$

$d\vec{a} = da \hat{n}$ da : area of the element
 $\hat{n}$: unit vector

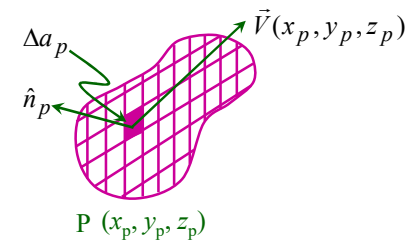
$\int_S \vec{V} \cdot d\vec{a}$ in general, depends on the choice of surface

There exists a class of vector field for which surface integral is independent of the choice of the surface

15

Vector Integration

Surface Integral:

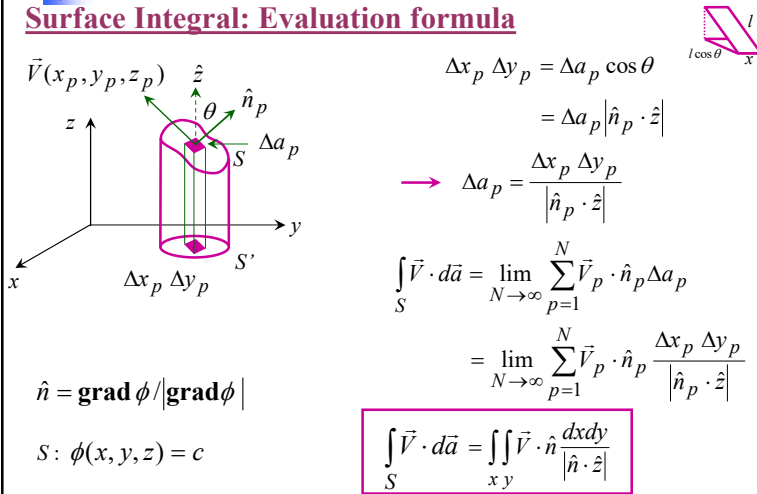


$$\int_S \vec{V} \cdot d\vec{a} = \lim_{N \rightarrow \infty} \sum_{p=1}^N \vec{V}(x_p, y_p, z_p) \cdot \Delta a_p \hat{n}_p$$

16

Vector Integration

Surface Integral: Evaluation formula



$\Delta x_p \Delta y_p = \Delta a_p \cos \theta$
 $= \Delta a_p |\hat{n}_p \cdot \hat{z}|$
 $\rightarrow \Delta a_p = \frac{\Delta x_p \Delta y_p}{|\hat{n}_p \cdot \hat{z}|}$

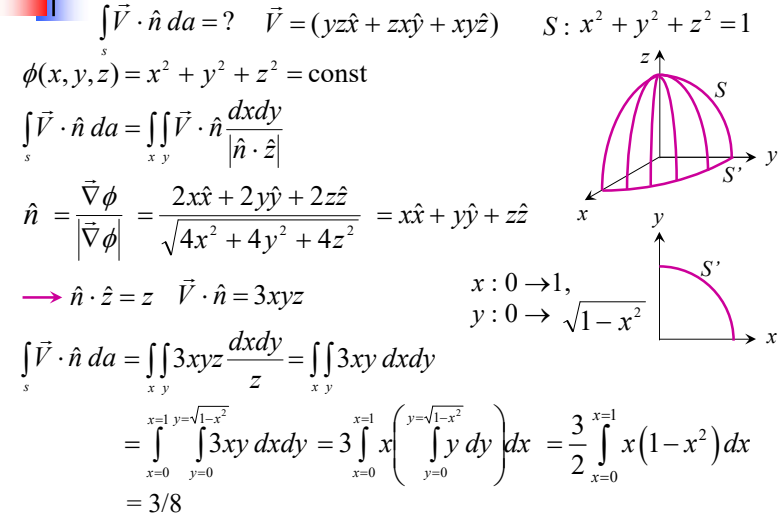
$\hat{n} = \frac{\nabla \phi}{|\nabla \phi|}$
 $S: \phi(x, y, z) = c$

$\int_S \vec{V} \cdot d\vec{a} = \lim_{N \rightarrow \infty} \sum_{p=1}^N \vec{V}_p \cdot \hat{n}_p \Delta a_p$
 $= \lim_{N \rightarrow \infty} \sum_{p=1}^N \vec{V}_p \cdot \hat{n}_p \frac{\Delta x_p \Delta y_p}{|\hat{n}_p \cdot \hat{z}|}$

$\int_S \vec{V} \cdot d\vec{a} = \iint_S \vec{V} \cdot \hat{n} \frac{dxdy}{|\hat{n} \cdot \hat{z}|}$

17

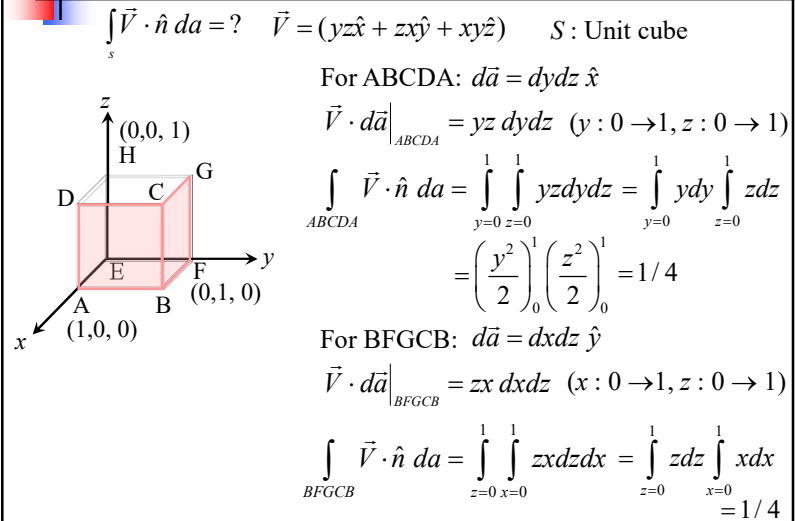
Surface Integration: Example 1



$\int_S \vec{V} \cdot \hat{n} da = ? \quad \vec{V} = (yz\hat{x} + zx\hat{y} + xy\hat{z}) \quad S: x^2 + y^2 + z^2 = 1$
 $\phi(x, y, z) = x^2 + y^2 + z^2 = \text{const}$
 $\int_S \vec{V} \cdot \hat{n} da = \iint_S \vec{V} \cdot \hat{n} \frac{dxdy}{|\hat{n} \cdot \hat{z}|}$
 $\hat{n} = \frac{\nabla \phi}{|\nabla \phi|} = \frac{2x\hat{x} + 2y\hat{y} + 2z\hat{z}}{\sqrt{4x^2 + 4y^2 + 4z^2}} = x\hat{x} + y\hat{y} + z\hat{z}$
 $\rightarrow \hat{n} \cdot \hat{z} = z \quad \vec{V} \cdot \hat{n} = 3xyz$
 $\int_S \vec{V} \cdot \hat{n} da = \iint_S 3xyz \frac{dxdy}{z} = \iint_S 3xy dxdy$
 $= \int_{x=0}^{x=1} \int_{y=0}^{y=\sqrt{1-x^2}} 3xy dxdy = 3 \int_{x=0}^{x=1} x \left(\int_{y=0}^{y=\sqrt{1-x^2}} y dy \right) dx = \frac{3}{2} \int_{x=0}^{x=1} x(1-x^2) dx$
 $= 3/8$

18

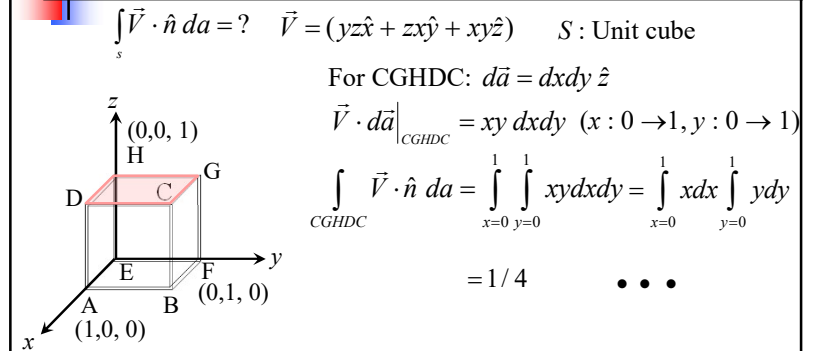
Surface Integration: Example 2



$\int_S \vec{V} \cdot \hat{n} da = ? \quad \vec{V} = (yz\hat{x} + zx\hat{y} + xy\hat{z}) \quad S: \text{Unit cube}$
 For ABCD: $d\vec{a} = dydz \hat{x}$
 $\vec{V} \cdot d\vec{a}|_{ABCD} = yz dydz \quad (y: 0 \rightarrow 1, z: 0 \rightarrow 1)$
 $\int_{ABCD} \vec{V} \cdot \hat{n} da = \int_{y=0}^1 \int_{z=0}^1 yz dydz = \int_{y=0}^1 y dy \int_{z=0}^1 z dz$
 $= \left(\frac{y^2}{2} \right)_0^1 \left(\frac{z^2}{2} \right)_0^1 = 1/4$
 For BFGCB: $d\vec{a} = dx dz \hat{y}$
 $\vec{V} \cdot d\vec{a}|_{BFGCB} = zx dx dz \quad (x: 0 \rightarrow 1, z: 0 \rightarrow 1)$
 $\int_{BFGCB} \vec{V} \cdot \hat{n} da = \int_{z=0}^1 \int_{x=0}^1 zx dx dz = \int_{z=0}^1 z dz \int_{x=0}^1 x dx$
 $= 1/4$

19

Surface Integration: Example 2



$\int_S \vec{V} \cdot \hat{n} da = ? \quad \vec{V} = (yz\hat{x} + zx\hat{y} + xy\hat{z}) \quad S: \text{Unit cube}$
 For CGHDC: $d\vec{a} = dx dy \hat{z}$
 $\vec{V} \cdot d\vec{a}|_{CGHDC} = xy dx dy \quad (x: 0 \rightarrow 1, y: 0 \rightarrow 1)$
 $\int_{CGHDC} \vec{V} \cdot \hat{n} da = \int_{x=0}^1 \int_{y=0}^1 xy dx dy = \int_{x=0}^1 x dx \int_{y=0}^1 y dy$
 $= 1/4 \quad \dots$

20

Volume Integration

$$\int_V \phi dV$$

$$\int_V \vec{F} dV$$

$$dV = dx dy dz$$



Essentially a triple integral

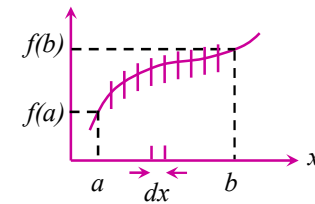
21

Fundamental Theorem of Calculus

$$\frac{df}{dx} = F(x)$$

$$\text{FTC} \quad \int_a^b \frac{df}{dx} dx = f(b) - f(a)$$

$\frac{df}{dx} dx = df$ infinitesimal change in f with infinitesimal change in x



22

Fundamental Theorem ...

$$\int_a^b \frac{df}{dx} dx = f(b) - f(a)$$

$$\phi(x, y, z)$$

$$d\phi = \vec{\nabla} \phi \cdot d\vec{l}$$

$$d\phi_1 = \vec{\nabla} \phi_1 \cdot d\vec{l}_1 = \phi(1) - \phi(a)$$

$$d\phi_2 = \vec{\nabla} \phi_2 \cdot d\vec{l}_2 = \phi(2) - \phi(1)$$

$$d\phi_N = \vec{\nabla} \phi_N \cdot d\vec{l}_N = \phi(b) - \phi(N)$$

$$d\phi_1 + d\phi_2 + \dots + d\phi_N = \vec{\nabla} \phi_1 \cdot d\vec{l}_1 + \vec{\nabla} \phi_2 \cdot d\vec{l}_2 + \dots + \vec{\nabla} \phi_N \cdot d\vec{l}_N = \phi(b) - \phi(a)$$

with $N \rightarrow \infty$ $\boxed{\int_a^b \vec{\nabla} \phi \cdot d\vec{l} = \phi(b) - \phi(a)}$ $\longrightarrow \oint_L \vec{\nabla} \phi \cdot d\vec{l} = 0$

Fundamental Theorem of Gradient

23

Fundamental Theorem of Gradient: Example

Consider a scalar function $\phi = xy^2$ and points $a(0,0,0)$ & $b(2,1,0)$

Using FTOG

$$\int_a^b \vec{\nabla} \phi \cdot d\vec{l} = \phi(b) - \phi(a) = 2 \cdot 1^2 - 0 = 2$$

Path 1: For (i), $x: 0 \rightarrow 2, y=0, dy=dz=0 \quad d\vec{l} = dx \hat{x}$

$$\vec{\nabla} \phi \cdot d\vec{l} = (y^2 \hat{x} + 2yx\hat{y}) \cdot dx \hat{x} = y^2 dx \longrightarrow \int \vec{\nabla} \phi \cdot d\vec{l} = 0$$

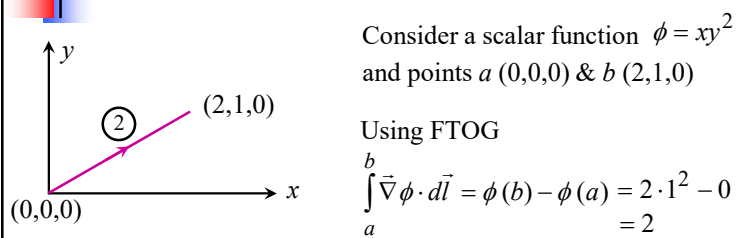
Path 1: For (ii), $x: 2, y: 0 \rightarrow 1, dx=dz=0 \quad d\vec{l} = dy \hat{y}$

$$\vec{\nabla} \phi \cdot d\vec{l} = (y^2 \hat{x} + 2yx\hat{y}) \cdot dy \hat{y} = 2yx dy \longrightarrow \int \vec{\nabla} \phi \cdot d\vec{l} = \int_0^1 4y dy = 2$$

$$\longrightarrow \int \vec{\nabla} \phi \cdot d\vec{l} = \int_i \vec{\nabla} \phi \cdot d\vec{l} + \int_{ii} \vec{\nabla} \phi \cdot d\vec{l} = 2$$

24

Fundamental Theorem of Gradient: Example



Path 2: $y = mx + c$, $m = 1/2$, $c = 0$, $y = x/2$, $dy = dx/2$

$$d\vec{l} = dx \hat{x} + dy \hat{y} \quad \vec{\nabla} \phi = (y^2 \hat{x} + 2yx \hat{y})$$

$$\vec{\nabla} \phi \cdot d\vec{l} = y^2 dx + 2yx dy = \frac{x^2}{4} dx + 2 \cdot \frac{x}{2} \cdot \frac{dx}{2} = \frac{3}{4} x^2 dx$$

$$\int_2 \vec{\nabla} \phi \cdot d\vec{l} = \int_0^2 \frac{3}{4} x^2 dx = 2$$

25

Fundamental Theorem of Divergence/Green/Gauss

$$\int_{\text{volume}} (\vec{\nabla} \cdot \vec{V}) d\tau = \oint_{\text{surface}} \vec{V} \cdot d\vec{a}$$

$$\int_{\text{volume}} (\vec{\nabla} \cdot \vec{V}) d\tau = \int_0^c \int_0^b \int_0^a \left(\frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z} \right) dx dy dz$$

$$\int_0^c \int_0^b \left(\int_0^a \frac{\partial V_y}{\partial y} dy \right)_{\text{const } x,z} dx dz$$

$$= \int_0^c \int_0^b [V_y(x, y=b, z) - V_y(x, y=0, z)] dx dz$$

$$= \int_0^c \int_0^b V_y(x, y=b, z) dx dz - \int_0^c \int_0^b V_y(x, y=0, z) dx dz$$

$$= \int_0^c \int_0^b \vec{V}(y=b) \cdot \hat{y} dx dz + \int_0^c \int_0^b \vec{V}(y=0) \cdot (-\hat{y} dx dz)$$

$$= \int_{S_1} \vec{V} \cdot d\vec{a} + \int_{S_2} \vec{V} \cdot d\vec{a}$$

26

Fundamental Theorem of Divergence/Green/Gauss

$$\int_{\text{volume}} (\vec{\nabla} \cdot \vec{V}) d\tau = \oint_{\text{surface}} \vec{V} \cdot d\vec{a}$$

$$\int_{\text{volume}} (\vec{\nabla} \cdot \vec{V}) d\tau = \int_0^c \int_0^b \int_0^a \left(\frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z} \right) dx dy dz$$

$$\int_0^c \int_0^b \left(\int_0^a \frac{\partial V_x}{\partial x} dx \right)_{\text{const } y,z} dy dz$$

$$= \int_0^c \int_0^b [V_x(x=a, y, z) - V_x(x=0, y, z)] dy dz$$

$$= \int_0^c \int_0^b V_x(x=a, y, z) dy dz - \int_0^c \int_0^b V_x(x=0, y, z) dy dz$$

$$= \int_0^c \int_0^b \vec{V}(x=a) \cdot \hat{x} dy dz + \int_0^c \int_0^b \vec{V}(x=0) \cdot (-\hat{x} dy dz)$$

$$= \int_{S_3} \vec{V} \cdot d\vec{a} + \int_{S_4} \vec{V} \cdot d\vec{a}$$

27

Fundamental Theorem of Divergence/Green/Gauss

$$\int_{\text{volume}} (\vec{\nabla} \cdot \vec{V}) d\tau = \oint_{\text{surface}} \vec{V} \cdot d\vec{a}$$

$$\int_{\text{volume}} (\vec{\nabla} \cdot \vec{V}) d\tau = \int_0^c \int_0^b \int_0^a \left(\frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z} \right) dx dy dz$$

$$\int_0^c \int_0^b \left(\int_0^a \frac{\partial V_z}{\partial z} dz \right)_{\text{const } x,y} dx dy$$

$$= \int_{S_5} \vec{V} \cdot d\vec{a} + \int_{S_6} \vec{V} \cdot d\vec{a}$$

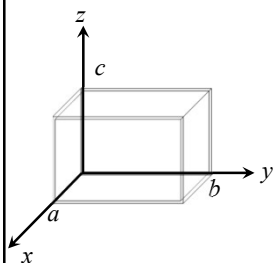
28

Fundamental Theorem of Divergence/Green/Gauss

$$\int_{\text{volume}} (\vec{\nabla} \cdot \vec{V}) d\tau = \oint_{\text{surface}} \vec{V} \cdot d\vec{a}$$

$$\int_{\text{volume}} (\vec{\nabla} \cdot \vec{V}) d\tau = \int_0^c \int_0^b \int_0^a \left(\frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z} \right) dx dy dz$$

$$= \int_{S_1} \vec{V} \cdot d\vec{a} + \int_{S_2} \vec{V} \cdot d\vec{a} + \int_{S_3} \vec{V} \cdot d\vec{a} + \int_{S_4} \vec{V} \cdot d\vec{a} + \int_{S_5} \vec{V} \cdot d\vec{a} + \int_{S_6} \vec{V} \cdot d\vec{a}$$

$$= \oint_S \vec{V} \cdot d\vec{a}$$


29

Fundamental Theorem of Divergence: Example

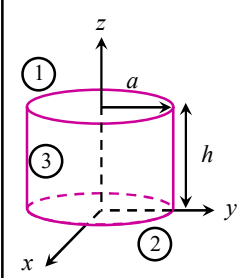
Consider a vector function $\vec{V} = \vec{r}$ Evaluate $\oint \vec{V} \cdot d\vec{a}$

$$\oint_S \vec{V} \cdot d\vec{a} = \int_{S_1} \vec{V} \cdot d\vec{a} + \int_{S_2} \vec{V} \cdot d\vec{a} + \int_{S_3} \vec{V} \cdot d\vec{a} = 3\pi a^2 h$$

For S_1 : $\hat{n} = \hat{z}$, $z = h$ $d\vec{a} = \hat{z} da$
 $\vec{V} \cdot d\vec{a} = z da = h da \rightarrow \int_{S_1} \vec{V} \cdot d\vec{a} = h \int da = h \pi a^2$

For S_2 : $\hat{n} = -\hat{z}$, $z = 0$ $d\vec{a} = -\hat{z} da$ $\vec{V} \cdot d\vec{a} = -z da = 0$
 $\rightarrow \int_{S_2} \vec{V} \cdot d\vec{a} = 0$

For S_3 : $\vec{n} = ?$ $\vec{n} = x\hat{x} + y\hat{y}$ $\hat{n} = \frac{x\hat{x} + y\hat{y}}{\sqrt{x^2 + y^2}} = \frac{1}{a}(x\hat{x} + y\hat{y})$
 $\int_{S_3} \vec{V} \cdot d\vec{a} = \frac{1}{a} \int (x^2 + y^2) da = \frac{a^2}{a} \text{ (area of the cylindrical surface) } = a \cdot 2\pi ah = 2\pi a^2 h$



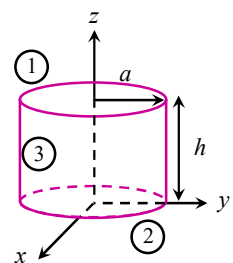
30

Fundamental Theorem of Divergence: Example

Consider a vector function $\vec{V} = \vec{r}$ Evaluate $\oint \vec{V} \cdot d\vec{a}$

$$\oint_S \vec{V} \cdot d\vec{a} = \int_{S_1} \vec{V} \cdot d\vec{a} + \int_{S_2} \vec{V} \cdot d\vec{a} + \int_{S_3} \vec{V} \cdot d\vec{a} = 3\pi a^2 h$$

Using FTOD:

$$\oint_S \vec{V} \cdot d\vec{a} = \int_V (\vec{\nabla} \cdot \vec{V}) d\tau = 3 \int_V d\tau = 3\pi a^2 h$$


31

Fundamental Theorem of Curl/Stokes Theorem

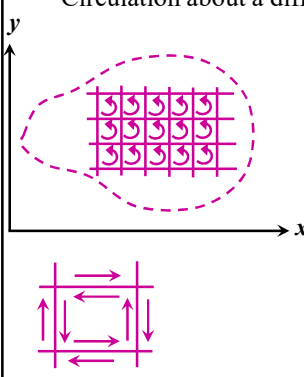
$$\int_S (\vec{\nabla} \times \vec{V}) \cdot d\vec{a} = \oint_L \vec{V} \cdot d\vec{l}$$

Circulation about a differential rectangle: $(\vec{\nabla} \times \vec{V})_z dx dy$

$$\text{or } \sum_{\text{four sides}} \vec{V} \cdot d\vec{l} = (\vec{\nabla} \times \vec{V})_z dx dy = (\vec{\nabla} \times \vec{V}) \cdot d\vec{a} \quad \because d\vec{a} = \hat{z} dx dy$$

$$\sum_N \sum_{\text{four sides}} \vec{V} \cdot d\vec{l} = \sum_N (\vec{\nabla} \times \vec{V}) \cdot d\vec{a}$$

$$\sum_{\text{exterior line segment}} \vec{V} \cdot d\vec{l} = \sum_{\text{rect}} (\vec{\nabla} \times \vec{V}) \cdot d\vec{a}$$

$$\oint_L \vec{V} \cdot d\vec{l} = \int_S (\vec{\nabla} \times \vec{V}) \cdot d\vec{a} \quad \text{FTOC}$$


32

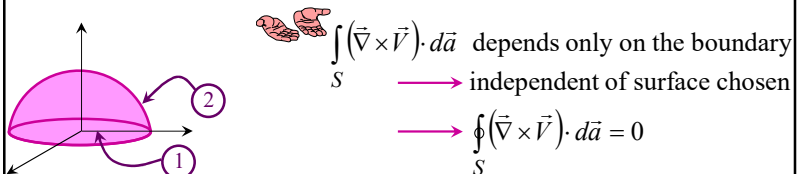
Fundamental Theorem of Curl/Stokes Theorem

$$\int_S (\vec{\nabla} \times \vec{V}) \cdot d\vec{a} = \oint_L \vec{V} \cdot d\vec{l}$$

Surface integral of a vector function, in general, depends critically on the surface we choose to integrate.

That's not true for the curls.

$\int_S (\vec{\nabla} \times \vec{V}) \cdot d\vec{a}$ is equal to the line integral of $\vec{V}$ around the boundary S which does not make any reference of the specific surface we choose



33

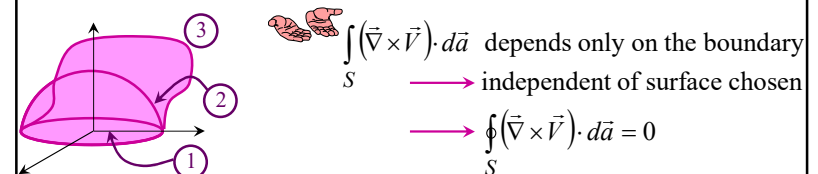
Fundamental Theorem of Curl/Stokes Theorem

$$\int_S (\vec{\nabla} \times \vec{V}) \cdot d\vec{a} = \oint_L \vec{V} \cdot d\vec{l}$$

Surface integral of a vector function, in general, depends critically on the surface we choose to integrate.

That's not true for the curls.

$\int_S (\vec{\nabla} \times \vec{V}) \cdot d\vec{a}$ is equal to the line integral of $\vec{V}$ around the boundary S which does not make any reference of the specific surface we choose



34

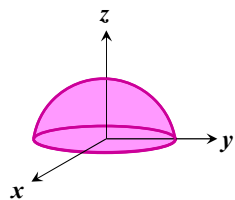
Fundamental Theorem of Curl: Example

Evaluate $\int_S (\vec{\nabla} \times \vec{V}) \cdot d\vec{a}$ over a hemisphere $x^2 + y^2 + z^2 = a^2; z \geq 0$

for $\vec{V} = 4y \hat{x} + x \hat{y} + 2z \hat{z}$

Boundary of the surface is $x^2 + y^2 = a^2$

One can choose any surface bounded by $x^2 + y^2 = a^2$



$$\hat{n} = \hat{z}$$

$$\vec{\nabla} \times \vec{V} = -3\hat{z}$$

$$(\vec{\nabla} \times \vec{V}) \cdot \hat{n} = -3$$

$$\int_S (\vec{\nabla} \times \vec{V}) \cdot d\vec{a} = -3 \int_S da = -3\pi a^2 \blacktriangleleft$$

35

Important Tool: Integration by Part

$$\frac{d}{dx}(fg) = f \frac{dg}{dx} + g \frac{df}{dx}$$

$$\int_a^b \frac{d}{dx}(fg) dx = \int_a^b f \left(\frac{dg}{dx} \right) dx + \int_a^b g \left(\frac{df}{dx} \right) dx = (fg)|_a^b$$

$$\int_a^b f \left(\frac{dg}{dx} \right) dx = - \int_a^b g \left(\frac{df}{dx} \right) dx + (fg)|_a^b$$

Example:

$$\begin{aligned} \int_0^\infty x e^{-x} dx &= ? & e^{-x} &= \frac{d}{dx}(-e^{-x}) & f(x) &= x & g(x) &= -e^{-x} & \frac{df}{dx} &= 1 \\ \int_0^\infty x e^{-x} dx &= \int_0^\infty x \frac{d}{dx}(-e^{-x}) dx = - \int_0^\infty (-e^{-x}) \frac{dx}{dx} dx + (-x e^{-x}) \Big|_0^\infty = \int_0^\infty e^{-x} dx - x e^{-x} \Big|_0^\infty \\ &= -e^{-x} \Big|_0^\infty - (0 - 0) = 1 \end{aligned}$$

36



Important Tool: Integration by Part

$$\frac{d}{dx}(fg) = f \frac{dg}{dx} + g \frac{df}{dx}$$

$$\int_a^b f \left(\frac{dg}{dx} \right) dx = - \int_a^b g \left(\frac{df}{dx} \right) dx + (fg) \Big|_a^b$$

$$\vec{\nabla} \cdot (f\vec{A}) = f (\vec{\nabla} \cdot \vec{A}) + (\vec{\nabla} f) \cdot \vec{A}$$

$$\int_{vol} \vec{\nabla} \cdot (f\vec{A}) d\tau = \int_{vol} f (\vec{\nabla} \cdot \vec{A}) d\tau + \int_{vol} (\vec{\nabla} f) \cdot \vec{A} d\tau = \oint_S f\vec{A} \cdot d\vec{a}$$

$$\int_{vol} f (\vec{\nabla} \cdot \vec{A}) d\tau = - \int_{vol} (\vec{\nabla} f) \cdot \vec{A} d\tau + \oint_S f\vec{A} \cdot d\vec{a}$$